

Loss Aversion in Ambiguous Decision Making

Group 2 | Discussion Section 305

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Abstract (Jacob)

Ambiguity aversion is the common preference for known situations, commonly measured by the Ellsberg paradox, in which people are more likely to pick a known probability gamble than an unknown probability of the same average value. Loss aversion is also well documented, in that the negative effects of a loss are felt more strongly than the positive effects of an equivalent gain. We hypothesized that people would be more likely to favor a known gamble when the situation was framed as a loss compared to when framed as a gain. Our behavioral measure was a web app which presented the classic Ellsberg paradox, a jar of 30 red balls and 60 balls that are either black or yellow, then tasked with choosing a red (known) or black (unknown) ball. If the correct ball was chosen, participants would either win or lose 100 dollars. We expect more participants to pick the known gamble in the loss framing condition. This may correlate with more activation of the amygdala measured by fMRI, correlating with higher emotional salience in a loss correlating to less subjective value. This could be a significant finding in the integration of uncertainty and value representations in the brain.

Introduction (Emily)

Ambiguity aversion, which is avoiding making a choice with an unknown probability of success, has been studied using the Ellsberg Paradox since 1961. When given a risky choice and a choice with an unknown probability, people will often pick the risky choice as they know the probability of success, despite the success rate being low (Jia, 2020). While this concept has been studied in detail, less research has been done about how a

person's willingness to make a risky decision if they stand to lose something rather than gain something when the Ellsberg paradox is applied.

When presented with uncertainty, the brain gives more weight to new information that may be useful in coming to a decision (Jia, 2020). Additionally, there is a higher rate of learning when people are put in situations with uncertainty (Jia, 2020). Given a risky situation the brain has to weigh the benefits and potential consequences of either exploiting a known choice or exploring a new, untested choice to see if the outcome is better than the known choice (Osmont, 2023). The anterior cingulate cortex plays a valuable role in action selection while keeping reward expectations, prediction errors, and the value of exploring new choices in mind to better inform future decisions (Jia, 2020).

Although prior research has examined ambiguity aversion and risk aversion independently (Jia, 2020), not many have looked at how these two psychological processes work together when making a decision. In other words, it is still not clear whether a person's tendency to avoid ambiguity is stronger, weaker, or stays the same when the decision-making situation is framed in terms of losses instead of gains and which parts of the brain are involved in the decision making.

Loss aversion is one important aspect of the prospect theory, which was first developed by Daniel Kahneman and Amos Tversky, and states that people think of gains and losses relative to a reference point instead of in absolute terms. Loss aversion itself is the phenomenon of losses having a greater effect on a person's well-being as gains of the same caliber (Park, 2025). The prospect theory is a wider framework for assigning subjective values to losses and gains. It uses the KT model to evaluate the relationship between the objective value and the subjective value of the gain or loss in the scenario (Park, 2025). The KT model utilizes the concepts of loss aversion, reference dependence, and diminishing sensitivity to infer how people will change the magnitude of the subjective value they assign to the reward they stand to gain, or what they stand to lose (Park, 2025).

Decision making in the face of uncertainty primarily takes place in the dorsal lateral prefrontal cortex and orbital frontal cortex. More specifically, it was found that generally the dorsal lateral prefrontal cortex impacts risk taking while the orbital frontal cortex is more involved with ambiguity (Hu, 2023). Khaleghi et al. (2020) however, indicated that the specific neural mechanisms involved in risk-taking behavior varied depending on what task researchers had subjects do. This finding signifies that different brain areas would have varying activation for different tasks despite the fact that the tasks were all meant to measure risk-taking behavior and decision making under uncertainty. It was found that stimulation to the left dorsal lateral prefrontal cortex led to more stable decision making under ambiguous circumstances and led to less loss aversion among subjects (Hu, 2023).

In our study we aim to test whether the Ellsberg Paradox and ambiguity aversion still applies when participants are presented with a risky choice resulting in them losing something and an ambiguous choice with an unknown probability of losing something. By comparing the standard version of the Ellsberg paradox with a control group to a group given a modified Ellsberg Paradox where the gamble would result in a loss, we can test if there is a difference in behavior given how the gamble is framed.

We hypothesize that people will avoid ambiguous situations at a higher rate when gambles are framed as a loss rather than a gain. Specifically, the proportion of choosing the known option in an Ellsberg paradox situation will be lower for the standard model when the gamble is to win a reward. When the situation is reframed as losing a reward, the proportion picking the known option will increase, reflecting increased loss aversion in ambiguous situations.

Methods (Yifan)

Participants

Participants will be recruited from the University of Wisconsin–Madison, approximately 100 to 120 undergraduate students through email. All participants will be between the ages of 18 and 22. The recruitment email will provide a brief description of the study, outline any potential risks, and attach a link to the online experiment. Participation will be conducted anonymously. While participants will be required to enter a Participant ID during the experiment, However, the ID does not need to have any relation to one's personal information.

Materials

This experiment will use a web-based online application developed by the experimenters, designed to present the "Ellsberg Paradox." Participants will be placed in a scenario involving a container holding 90 balls: 30 red balls (with a known probability of $1/3$) and 60 black or yellow balls (with an unknown specific ratio, thereby introducing ambiguity regarding their probabilities).

The experiment represents a modification of the standard "Ellsberg Paradox" paradigm, wherein participants make their choices after being informed of the potential outcome—specifically, whether the result will be a gain or a loss. Participants will be asked to make choices under two distinct scenarios: in the first scenario, they are informed that the outcome will result in a loss of \$100; in the second scenario, they are informed that the outcome will result in a gain of \$100. The probability of encountering either scenario is randomized.

Design

This study employs a between-subjects design, in which participants will be randomly assigned to one of two "framing conditions":

Gain Framing Condition: Participants are informed that if the color of the ball they select matches a designated color, they will win \$100.

Loss Framing Condition: Participants are informed that if the color of the selected ball matches a specified color, they will incur a loss of \$100.

This goal is to examine the impact of the "framing effect"—specifically, the distinction between a gain scenario and a loss scenario—on the phenomenon of "ambiguity aversion."

Procedure

Participants will complete the experiment mentioned above, choosing between a red ball (representing a known probability) and a black ball (representing an uncertain probability) under conditions where the outcome involves either winning \$100 or losing \$100. Participants simply need to click on one of the options to register their choice. The backend database linked to the webpage will record the participants' selections, involving the following data:

Participant's Choice (Known Option vs. Ambiguous Option)

Experimental Condition (Gain vs. Loss)

Reaction Time (in milliseconds)

Reaction Timestamp

Results (Jacob)

We expected to observe a significant difference in the proportion picking the known gamble between the two groups. Specifically, we predicted that the loss framing group will have a higher proportion of picking the known gamble compared to the win framing group. Based on the classic Ellsberg paradox which has a gain frame, a majority of people will pick the known option anyway (Segal, 1987.) However, in the loss frame, we expected an even greater majority picking the known option, close to every participant.

In our final results, we were only able to collect data from 17 different participants in all, with 7 in the gain frame and 10 in the loss frame. This is a big limitation with our data set. Our methods limited the number of trials for each participant to one, so the sample size is not large enough to make strong conclusions.

Condition	Known Gamble	Ambiguous Gamble	Total
Grain Frame	4	3	7
Loss Frame	9	1	10
Total	13	4	17

Table 1 - Summary of participant choices by framing, with 17 participants total.

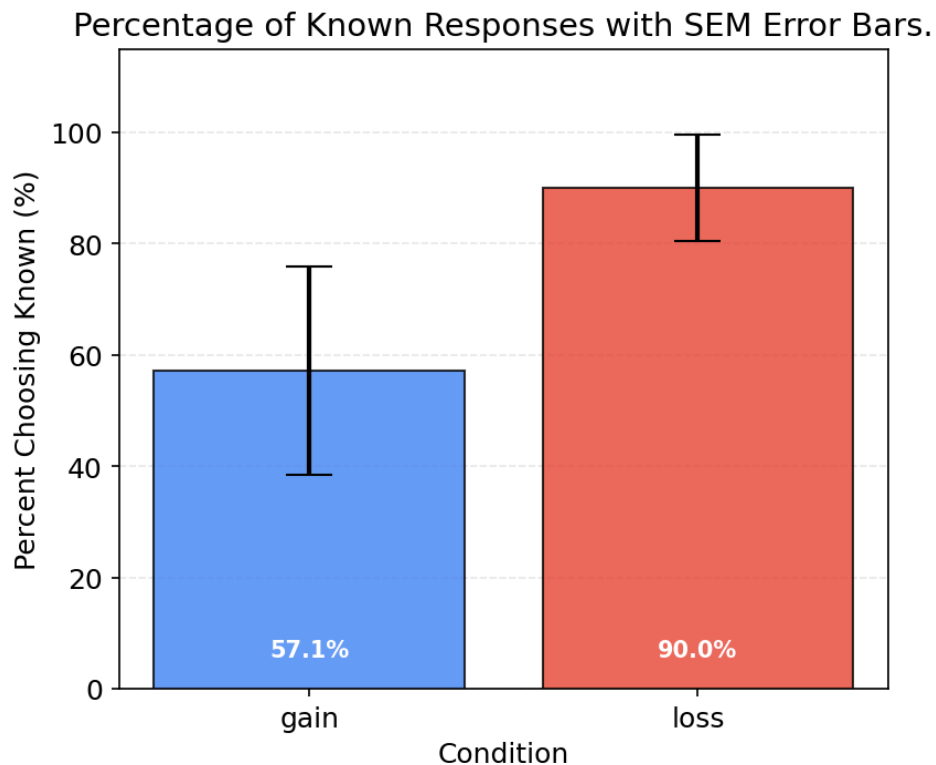


Figure 1 - Results for proportion choosing the known option in gain and loss frames. The loss frame has a higher proportion choosing the known option, based on non overlapping SEM error bars.

Our results show promise using SEM error bars, but we also ran a chi square test for independence to compare the two groups. With 1 degree of freedom and a chi square value of 0.982, the p value came out to be 0.322, which is not strongly statistically significant. To further analyze, the 95% confidence intervals (20.5% - 93.8% for the gain

frame, 71.4%, 100.0% for the loss frame) also overlap, indicating no significant difference. However, Cohen's h value of 0.784 to compare proportions indicates a strong effect size.

There is some support for our hypothesis based on an initial graphical analysis and a large effect size, but due to a small sample size, chi square and confidence interval tests show that the data is not strong enough to say it is statistically significant. This blend of statistical results indicates that an effect may be present, but there is not a large enough sample size to be extremely confident yet. However, based on the current data, a stronger trend may emerge consistent with the idea that people will be more averse to ambiguous situations in a loss frame.

Neuroscience Extensions (Yifan, Jacob)

Based on the data obtained from our experiment, the results so far are consistent with our predictions: when individuals are placed in a “loss frame” situation, they exhibit a stronger tendency to avoid uncertain risks.

In this study, we will use functional magnetic resonance imaging (fMRI) to monitor the amygdala of participants' brains, with a particular focus on changes in blood oxygen level-dependent (BOLD) signals within this region (Logothetis & Pfeuffer, 2004). This approach is based on previous research on the amygdala; therefore, we hypothesize that the amygdala is more likely to exhibit increased activity when individuals face situations dominated by negative emotions (Williams et al., 2006). In our case, it would be loss framing.

Based on our predictions of amygdala activity, we can hypothesize that in loss frames, more active amygdala activation may amplify and intensify people's perception of negative emotions, leading them to make safer, more certain choices. Conversely, in gain frames, where there are no negative emotions, the amygdala is inactive. In this

lower activation state, people are more tolerant of uncertain probabilities, leading them to make choices with uncertain probabilities.

A different brain region that may have the opposite correlation is the ventromedial prefrontal cortex. This region is involved in integration of subjective value in the brain (Bartra et al., 2013). Previous research has demonstrated that part of the integration aspect involves loss of subjective value in cases of uncertainty, another example of ambiguous aversion in the brain (Levy et al., 2010).

Putting these ideas together, using BOLD fMRI signals in the vmPFC could show a decrease in activation in the area for the loss frame group relative to the gain frame group. In addition, with more subjects, this decrease may be less pronounced in the individuals in the loss frame group that still go with the ambiguous option. This could help to explain the individual differences in risk tolerance under uncertain conditions.

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